Multimodal Assistant for Visually Impaired People

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The development of assisted technologies for visually impaired (VI) people has been accelerated in the past few years by the advancement of artificial intelligence, in particular, deep learning models; they focus on providing one or several functions such as obstacle detection and navigation, both indoors and outdoors, scene description, and text transcribing and reading. This paper presents a pipeline capable of offering all these functions, where it is able to process images and audio as inputs and output audio to engage conversationally with the user. We first process the image input through a fine-tuned YOLOv9 model for object detection and a MiDaS model for depth estimation. We integrated the outputs from both models to construct a structured shared visual context. This context, along with the original image and a transcribed audio query, is provided as input to the Gemma 3 model, which serves as the Vision-Language Question Answering (VQA) system. Finally, the textual answer generated by the model is converted into speech using a text-to-speech (TTS) system.
You can see our demo here, our video here and our GitHub here.

1 Introduction

According to the International Agency for the Prevention of Blindness, an estimated 43 million people live with blindness and 295 million with moderate to severe visual impairment worldwide [1]. Individuals with visual impairments face significant challenges in performing everyday tasks independently. Recent advancements in deep learning, mobile computing, and cloud-based computer vision services have accelerated the development of assistive technologies aimed at addressing these challenges [2]. These technologies often focus on specific capabilities such as object detection [3], distance estimation [4], text recognition [5], and scene description [6]. Our work proposes a unified pipeline that enables users to access multiple assistive functions, including object detection, distance estimation, text reading, and scene understanding, through a single, accurate, and comprehensive multimodal tool designed to support visually impaired individuals in real-world scenarios.

2 Related Work

Recent work by Yang et al. introduced VIAssist [7], which adapts multimodal large language models (MLLMs) to support users with visual impairments. Their system leverages MLLMs such as LLaVA-1.5 and Qwen-VL-Chat to answer visual questions posed by visually impaired (VI) users. Inspired by this approach, our proposed pipeline builds upon the MLLM framework by incorporating a structured shared visual context. We obtain the shared visual context by combining outputs from object detection and depth estimation as an additional input to the MLLM. In our case, we utilize the recently released Gemma 3 model as the core VQA engine.

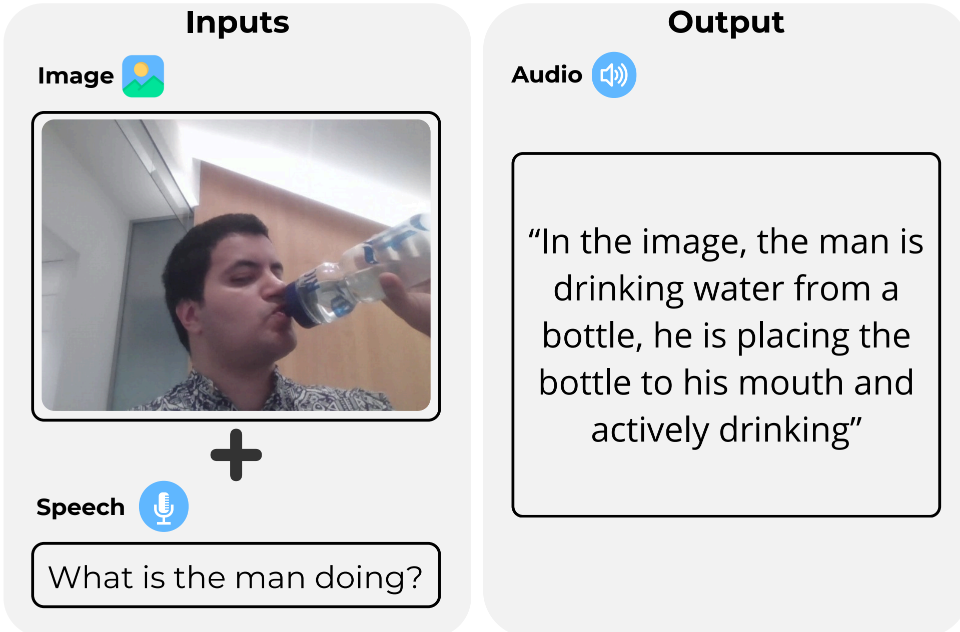


Figure 1: In this figure we show an example of our pipeline’s inputs and resultant output.

3 Methodology

Our proposed pipeline (Figure 2) integrates visual and audio inputs to support conversational interaction for visually impaired users. Three core modules process the image input: First, the **Object Detection** module, a fine-tuned YOLOv9 model [8] identifies and localizes salient objects in the scene. Second, the **Depth Estimation** module, a MiDaS model [9] estimates relative depth, generating a spatial layout of the visual field, and third, the **Vision-Language Question Answering (VQA)** module, where Gemma-3 4B IT model [10] processes visual inputs and text queries to produce natural language answers.

Object detection and depth estimation outputs are merged into a structured textual summary called the shared visual context. This summary provides explicit scene-level understanding (e.g., object types, positions, and spatial relationships) and is passed along with the original image to the VQA module to enhance reasoning.

Concurrently, Whisper [11], an automatic speech recognition (ASR) model, transcribes the spoken queries. The resulting text is combined with the image and shared visual context as input to the VQA module, which generates a response in natural language. This response is converted into audio output via a Text-to-Speech module [12] to enable seamless interaction.

The key innovation of our pipeline is incorporating the shared visual context as an intermediate representation, enabling more informed and explainable answers from the VQA module, especially in visually complex or low-quality scenes.

3.1 Object Detection using a Fine-tuned YOLOv9 model

YOLOv9 is an object detection algorithm renowned for its accuracy and real-time performance [6]. It operates by dividing the input image into a grid of cells and predicting bounding boxes and class probabilities for objects within each cell. YOLOv9 introduced the Programmable Gradient Information (PGI) method, which improves gradient flow during backpropagation, helping the optimizer better learn complex patterns without suffering from vanishing or exploding gradients [8]. We finetuned the model using the VizWiz dataset [13] to improve its performance further (See Appendix B).

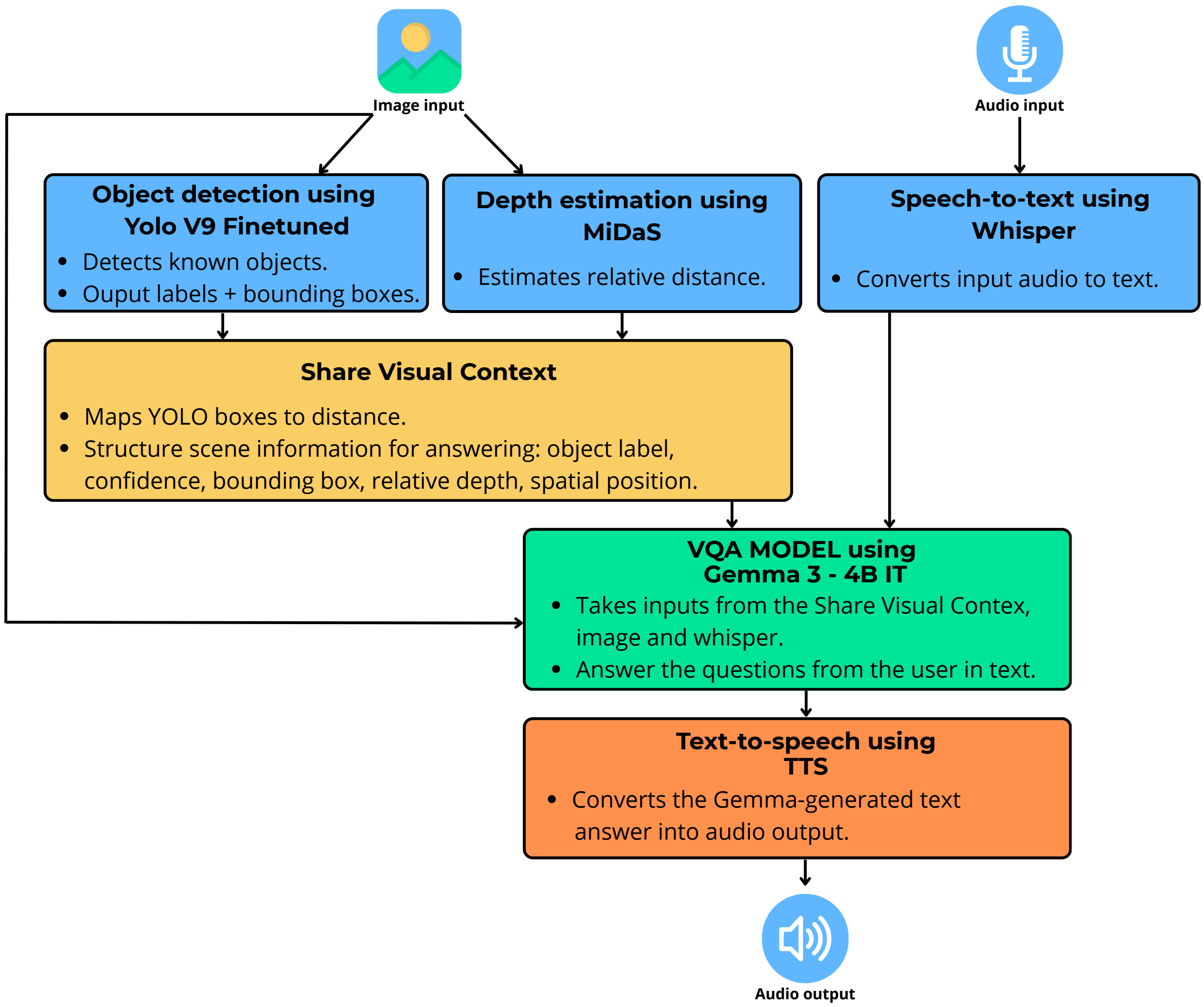


Figure 2: The figure details our pipeline components and workflow

3.2 Depth Estimation using MiDaS

The MiDaS model leverages an encoder-decoder architecture, where the encoder extracts high-level features from the input image, and the decoder predicts dense depth maps. By training on a mixture of datasets and employing scale-invariant loss functions, MiDaS achieves state-of-the-art performance in zero-shot cross-dataset transfer scenarios [9].

3.3 Shared Visual Context

The shared visual context comprises a list of detected objects, where each entry contains an object label, bounding box coordinates, a detection confidence score, the average depth of the MiDaS object region, and an estimated position based on the horizontal image location. It is an intermediate knowledge layer, enabling the Gemma 3 VQA model to ground relative distance and position questions (See Appendix C).

3.4 Speech-to-Text component based on Whisper

Whisper is an open-source automatic speech recognition (ASR) system developed by OpenAI. It converts spoken audio into text. Unlike traditional ASR systems that rely on separate components (e.g., acoustic model, language model), Whisper is a single, end-to-end transformer-based model trained on 680,000 hours of multilingual and multitask supervised data. We use the miniature version with 244M parameters [11].

3.5 The Gemma-3 4B IT model

Gemma 3 is a multimodal model recently added to the Gemma family of lightweight open models, ranging in scale from 1 to 27 billion parameters. This version introduces vision-understanding abilities, more comprehensive coverage of languages, and a longer context. The Gemma 3 models are trained with distillation and achieve superior performance to Gemma 2 for both pre-trained and instruction-finetuned versions. We use the Gemma3-4B-IT version that is on par with the Gemma2-27B-IT [10].

3.6 Text-to-speech using TTS

We use the Text-to-Speech (TTS) model from the SpeechBrain toolkit. It uses the Tacotron 2 architecture and is trained on the LJSpeech dataset, a popular English speech corpus [12].

4 Experiments

We comprehensively evaluated our visual assistant pipeline by testing combinations of its core capabilities—obstacle detection, object recognition, navigation, scene description, relative distance estimation, and text reading—within a conversational framework designed for visually impaired users. To simulate real-world challenges, we used images of varying difficulty, including those with poor lighting, occlusions, and motion blur (see Appendix A), thereby assessing the pipeline's robustness across diverse visual conditions.

In the second phase, we evaluated the pipeline on two tasks from the VizWiz Grand Challenge datasets: Image Captioning and Visual Question Answering (VQA) [13]. These datasets consist of images captured by blind users via a mobile application, often accompanied by spoken questions. We tested model performance with and without shared visual context (i.e., generated scene descriptions), using subsets of the validation splits for each task. For the VizWiz-VQA dataset, our evaluation focused on answer prediction and determining whether the question was answerable. In contrast, the VizWiz-Captions dataset requires generating accurate and coherent captions for user-taken images, thus testing the model's descriptive abilities [13]. Together, these tasks allowed us to assess the visual assistant's functional and generative aspects.

5 Results

We showed that each of the models' capabilities mentioned above is present in the pipeline, with different combinations and difficulties with images. (See examples in Appendix A). Here, we include some results from our second experiment.

5.1 Visual Question Answering

We evaluated our visual assistant pipeline's ability to distinguish answerable from unanswerable visual questions using the Answerability Average Precision (AP) metric, focusing on the VizWiz dataset, which contains challenging real-world images (e.g., blurry, poorly framed, low light). Accurately detecting unanswerable queries is crucial to avoid misleading outputs.

We tested the pipeline with and without the shared visual context, which incorporates scene information from object detection and depth estimation. The pipeline achieved an AP of 0.9100 with the shared visual context and 0.9009 without it, indicating strong overall performance with a slight improvement from scene-level context. It is important to note, however, that ground truth answerability labels are based on a rule-based heuristic from the dataset itself: a question is labeled as answerable if at least one of the ten crowd-sourced answers is marked with high confidence and is not "unanswerable." While this reflects human judgment, it does not necessarily capture the model's internal uncertainty.

5.2 Image Captioning

The BERT F1 metric assesses the semantic similarity between generated captions and ground-truth captions by computing the F1 score based on token overlap in BERT's embedding space. Higher values indicate greater semantic alignment between predicted and reference captions. In this experiment, the model incorporating the shared visual context achieved an average BERT F1 score of 85, slightly outperforming the baseline model without SVC, which scored 84. This result suggests that integrating the shared visual context contributes to marginally improved semantic accuracy in the generated image captions. Although the improvement is modest, the result supports the hypothesis that providing additional structured visual information (via the shared visual context) can enhance the model's ability to generate more semantically accurate descriptions.

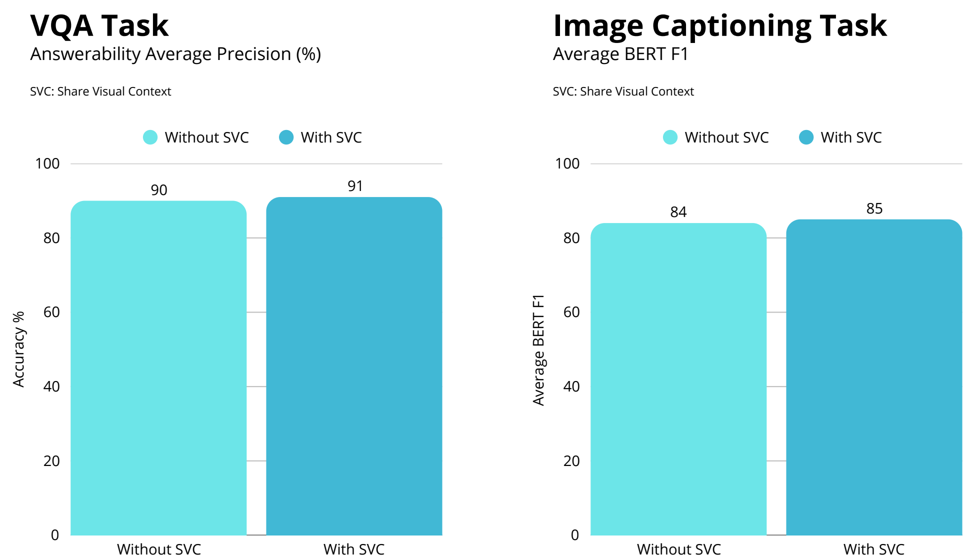


Figure 3: Comparison of model performance before and after the inclusion of the shared visual context. This shows that using the shared visual context slightly improves the performance of the model in

6 Conclusion

In this paper, we presented the pipeline that is designed to assist visually impaired users. Our major contribution is the addition of a share visual context in combination with a powerful multimodal large language model (MLLM) such as Gemma 3. The integration of these novel components allows the model to reason over image content in a context-aware manner, improving accuracy, interpretability, and consistency across interactions. Our experiments demonstrate that incorporating visual context slightly improves response quality. The use of the Gemma model, guided by task-specific prompting, ensures answers remain focused and user-appropriate.

Although this pipeline was able to produce some promising results, future work is needed to achieve better performance. Looking ahead, several promising directions remain. Fine-tuning the Gemma model on VQA-specific dialogue could improve its alignment with accessibility goals. Expanding the shared visual context to include scene attributes (like lighting conditions or crowd density) could further enhance informativeness. Finally, leveraging larger or more specialized MLLMs could boost answer accuracy and robustness in more complex or visually dense environments.

Our work lays a strong foundation for accessible, real-time visual assistance and opens the door to more profound multimodal understanding for visually impaired users.

Appendix A

Example 1

Test capabilities: Obstacle detection and navigation.
Difficulty: None. The image has no issues.

Is there any obstruction?

Yes, there are obstructions. The image shows a cross road with painted lines indicating where to walk.

Which way could I go?

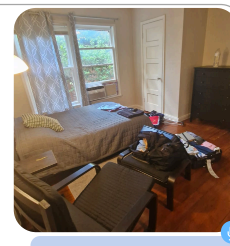
Based on the image, there are cars on both the left and the right side of the road. There are more cars on the left. To help to decide which way to go, I need more information. Do you want to go towards the trafic light on the left or away from it to the right?.

I want to cross the street

Okay, let me analyze the image, there is a cross walk directly in front of you. There are several cars on the street, some on the left, some on the right. There is a traffic light on the right side of the street. You should proceed cautiously, and wait for the traffic light to turn green before crossing. Would you like me to describe anything else in the scene?

Example 2

Test capabilities: Scene description and relative distance measurement.
Difficulty: None. The image has no issues.



Describe what you see

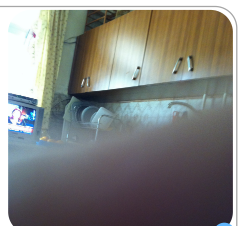
The image shows a bedroom. The image shows a bedroom. Furniture: There's a bed centered in the frame. To the left of the bed is a black chair. There's also a dark dresser to the right of the bed. A second, less distinct chair is visible in the center. Items: You can see a suitcase, a book, and a laptop are scattered around the bed. There's a backpack nearby as well. Windows: There's a window with curtains in the background.
Walls & Flooring: The walls appear to be a neutral beige color, and the floor is made of wood planks.

Which thing is closer to me?

Based on the image, the chair closest to you is the one on the left, located at 25.0 units.

Example 3

Test capabilities: Scene description.
Difficulty: Camera is obstructed.



Where am I?

Based on the image, you appear to be in a kitchen. Specifically, there's a sink, cabinets, and a television, which are all common features of a kitchen. Would you like me to describe any particular element in more detail?

What is in front of me?

Based on the image, it appears there is a bowl in front of you, located approximately 2.3 units to your center.

Where is the TV?

I can see a TV. It's located on the left side of the frame, roughly in the center.

Example 4

Test capabilities: Text reading.
Difficulty: Image is ill-framed.

What is this?

The item in focus is a container of "Lunch Maid Finely Sliced Oven Roasted Tomatoes." It's a blue plastic container with green text and a graphic of tomatoes on it.

Is there more details?

I can see a container of "Lunch Maid Finely Sliced Oven Roasted Tomatoes." It highlights that the product is "99% Fat Free" and "Reusable." It also states "Keep Refrigerated" and the net weight is "9 oz (255g).

It has a price on it?

I cannot see any price information on the packaging of the lunch meat in the image. The image primarily focuses on the product itself and its labeling.

Figure 4: Example of experiments testing different capabilities of the pipeline.

Appendix B

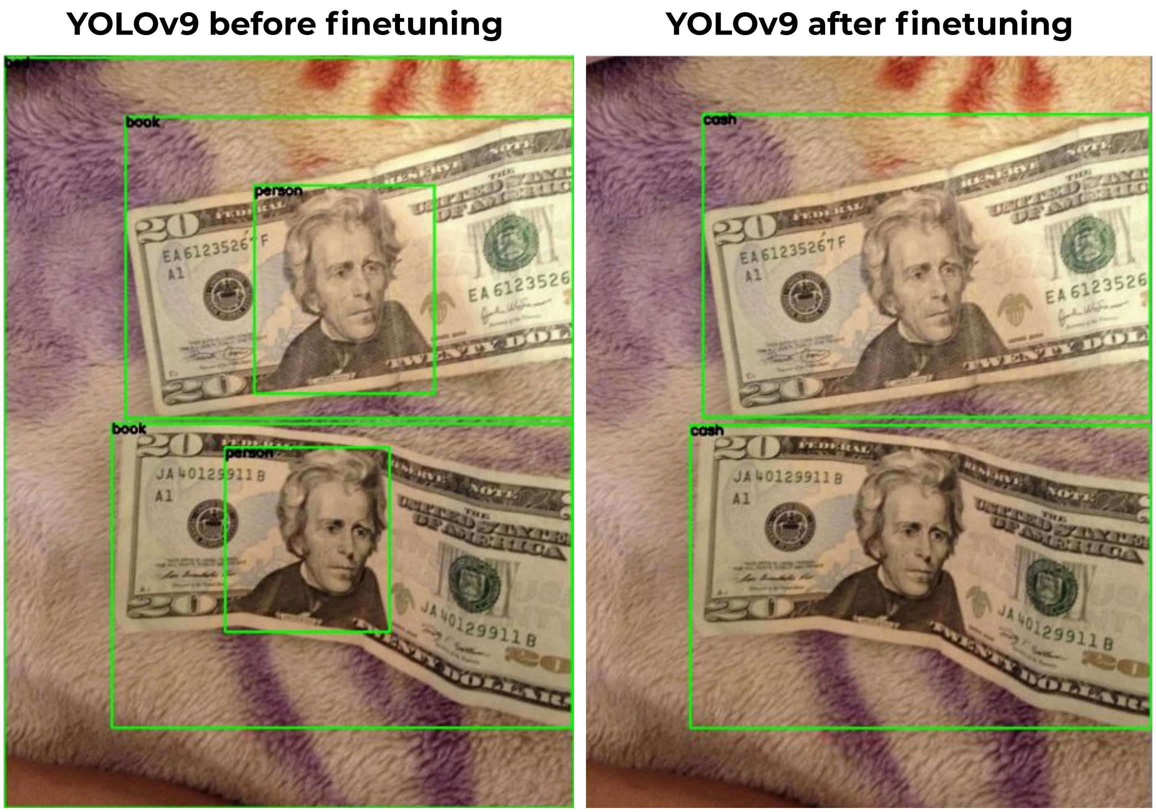
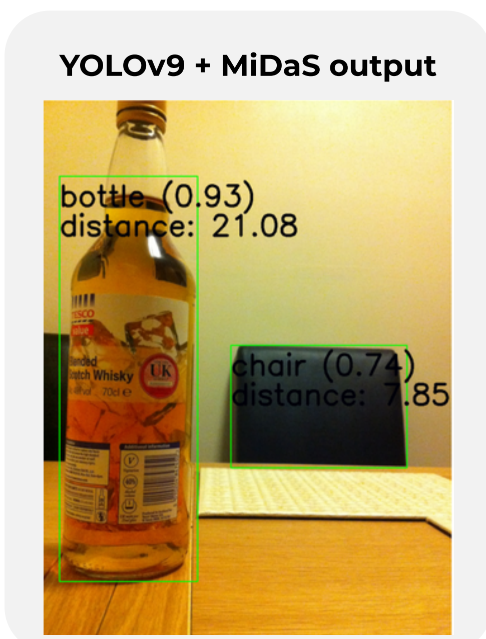


Figure 5: Results on VQA Task and Image Captioning Task on VizWiz dataset

Appendix C



Share Visual Context

[{'avg_depth': 21.08,
 'bbox': [39, 179, 366, 1138],
 'confidence': 0.92,
 'label': 'bottle',
 'position': 'left'},
{'avg_depth': 7.85,
 'bbox': [445, 579, 862, 869],
 'confidence': 0.74,
 'label': 'chair',
 'position': 'right'},
{'avg_depth': 21.84,
 'bbox': [0, 847, 968, 1293],
 'confidence': 0.49,
 'label': 'dining table',
 'position': 'center'}]

Figure 6: On the left, output of YOLOv9 and MiDaS. On the right, an example of the Shared Visual Context

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